Data analyst review and sign-off - 2.7.x

Data analyst review and sign-off - 2.7.x upgrade

Date: 12 November 2026

Scope: development database `labdb` on `172.16.16.44`, release `2.7.0`, schema head `0014_performance_indexes`.

Reviewer role: Data Analyst (upgrade gate).

Authority

This review is the data-quality gate for the 2.7.x upgrade. The analyst signs off on schema integrity, referential integrity, row/column sanity, and metric consistency. A conditional sign-off permits the development deployment to continue; it does not authorise production promotion while conditions remain open. The production gate is `docs/PROD-DEPLOYMENT-VALIDATION-MATRIX.md`.

Checks performed

Check	Method	Result
Referential integrity	Every foreign key in `public` checked for orphan child rows	PASS - 0 orphans
Critical nulls	`users`, `tracks`, `levels`, `labs`, `lab_sessions`, `lab_completions`, `courses`, `achievements`, `platform_settings`, `revoked_tokens`	PASS - 0 unexpected nulls
Uniqueness	Usernames, lower(email), lab slugs, (track, level)	PASS - 0 duplicates
Numeric sanity	Negative durations, counters, sort orders; future timestamps	PASS - 0
Status domains	`lab_sessions.status`	PASS - stopped/expired/error only
Report integrity	`report_exports.sha256`, `signature`, `report_id`	PASS - 0 missing
Lab reconciliation	DB slugs vs repository `lab.yaml` files	PASS - 178 = 169 repo + 9 intentional `demotest-*`
Ledger	`data_integrity_ledger` rows and attribution	WARN - 6,928 of 7,785 unattributed
Schema	47 tables, 14 Alembic revisions, triggers and indexes present	PASS
Image pipeline	Avatar/logo validation tests	PASS - 7/7

Findings

F1 - Ledger attribution is noisy (warning, not a blocker)

89% of ledger rows have no actor. The top tables are `labs`, `lab_completions`, `lab_sessions`, `users`, `achievements`, and `levels`. This is consistent with seed, discovery, migration, and scheduler writes that run without an authenticated request context. The rows are not evidence of tampering, but the review queue becomes hard to use.

Recommendation: set an explicit system actor for seed/discovery/scheduler paths, or add a `system` actor class to the ledger. Add a monthly review of unattributed rows and a retention window (for example 24 months) so the ledger does not grow without bound.

F2 - Demo data is mixed with curriculum data (accepted)

Nine `demotest-*` labs exist only in the database. They are intentional test fixtures and are not public. Keep them out of learner-facing metrics by filtering `visibility` and `is_active`; do not delete them while demo courses reference them.

F3 - Retention is not scheduled (warning)

`data_integrity_ledger`, `activity_events`, and `flag_attempts` grow without a retention job. At current volume this is not a risk, but the ledger grows with every write platform-wide.

Recommendation: add a scheduler job that archives or prunes rows older than the retention window and records the run in the ledger itself.

F4 - Metrics definitions

Lab counts differ by source: 169 repository labs, 178 database rows (9 demo), 49 non-public rows, 160 workbook pages. The README defines the metrics; dashboards should use the same definitions and label non-public/demo rows explicitly.

Sign-off

Decision: CONDITIONAL PASS for the development upgrade.

Conditions:

1. F1: system-actor attribution or an accepted-risk note before production.
2. F3: a scheduled retention job or an accepted-risk note before production.

Signed: Data Analyst Review (OpenCyberRange upgrade gate)

Date: 2026-11-12

Next review: before the production release candidate.